AgriSafeNet: Leveraging RGB-TIR UAV Remote Sensing Imagery for Transmission Line Detection in Smart Agriculture

Yichen Liu, Wenxuan Zhang, Xuanrui Chen, Zhao Ma, Junjie Ye, Guo Zheng, Lijie Wang, and Wei Hou

***Abstract*—In modern smart agriculture, the uncrewed aerial vehicles (UAVs) have emerged as indispensable low-altitude remote sensing platforms, particularly for precision irrigation and crop protection. While they enable the efficient acquisition of high-resolution spatial information and significantly enhance agricultural productivity, UAVs operating in complex environments face severe collision risks with power infrastructure, which poses a substantial threat to flight safety. Consequently, robust transmission line detection (TLD) is a critical prerequisite for ensuring safe autonomous UAV navigation. Despite the importance of this task, the existing detection methods often exhibit limitations in modeling fine-grained structural characteristics and fail to effectively integrate cross-modal contextual information. To address these challenges, this article proposes AgriSafeNet, a novel RGB-TIR TLD framework designed to bolster feature representation and multimodal fusion. Specifically, AgriSafeNet orchestrates efficient shallow-level cross-modal interaction through a multiscale feature decoupling and fusion module (MFDFM). Furthermore, it refines deep semantic integration via a spatial compression attention fusion module (SCAFM) and incorporates a context-guided feature extraction module (CGFEM) to capture intricate structural geometric details. This hierarchical design significantly improves the discriminability of transmission line (TL) features and the robustness of the model against environmental interference. Extensive experiments conducted on multiple benchmark datasets demonstrate that AgriSafeNet achieves the state-of-the-art performance, consistently outperforming the existing methods in complex agricultural scenarios.**

***Index Terms*—Low-altitude remote sensing, multimodal fusion, transmission line detection (TLD), uncrewed aerial vehicle (UAV).**

I. Introduction

ACCELERATED by rapid global economic development, smart agriculture [1], [2] has experienced widespread adoption and practical deployment. As key platforms for low-altitude remote sensing in precision farming, uncrewed aerial vehicles (UAVs) are increasingly utilized to acquire high-resolution spatial and spectral information for applications such as precision irrigation, pest and disease monitoring, crop protection, and environmental assessment [3], [4], [5]. Owing to the high maneuverability and efficient low-altitude data acquisition capabilities, UAVs play an essential role in improving agricultural productivity and enabling large-scale intelligent farmland management. However, as illustrated in Fig. 1, transmission lines (TLs) are widely distributed across agricultural environments, frequently crossing field ridges, rural roads, and plot boundaries, thereby forming critical obstacles within low-altitude remote sensing airspace. Due to their inherently slender and elongated structures, TLs pose significant collision risks to UAVs operating at low altitudes.

During agricultural field operations, UAVs typically execute missions at low altitudes, where flight trajectories frequently traverse areas densely populated by TLs and power towers. These structures constitute significant spatial obstacles that impede safe navigation. The widespread presence of such power infrastructure increases the probability of mid-air collisions. Such accidents not only cause substantial physical damage to UAV platforms but also compromise the integrity of critical power grids, potentially leading to localized electrical blackouts. Moreover, these collisions pose severe safety hazards to ground personnel and the surrounding ecological environment. Consequently, the precise detection and proactive avoidance of power facilities within complex agricultural scenarios represents a pivotal challenge for ensuring the safe and reliable deployment of autonomous UAVs.

As operational scenarios in smart agriculture become increasingly complex, UAV-based TLD faces substantial challenges. Within these environments, TLs typically exhibit ultrathin geometric structures and low visual contrast. The accurate perception of TLs is further complicated by adverse

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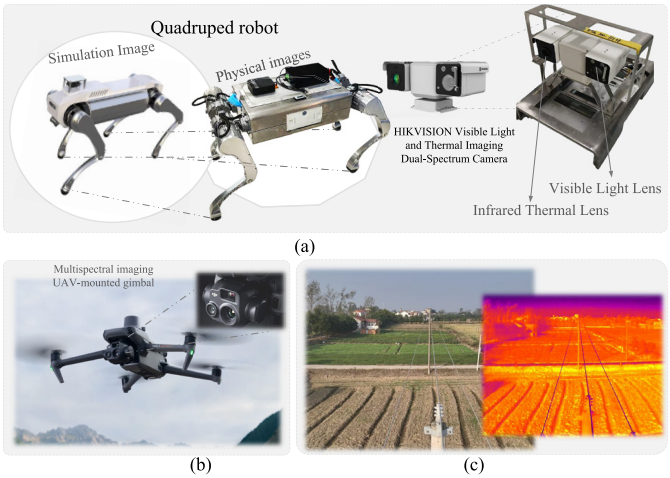


Fig. 1. Data acquisition platforms for the AgriSafe-TLD dataset. (a) Quadrupedal robot equipped with an RGB-TIR imaging dual-spectrum camera. (b) DJI Mavic 3T UAV. (c) Examples of captured RGB and TIR image pairs.

conditions, such as poor illumination, fog, and frequent partial occlusions. Consequently, these slender targets are easily confused with visually similar linear features, including irrigation canals and field boundaries. Because of the limited contextual information and semantic ambiguity inherent in single-modal visible-light imagery, conventional RGB-based methods struggle to accurately capture fine-grained structural characteristics, thereby failing to satisfy the stringent reliability requirements of UAV obstacle avoidance systems. To address these limitations, the fusion of RGB and thermal infrared imagery provides an effective solution. By exploiting the complementary properties of heterogeneous sensing modalities, the extraction of discriminative structural representations is substantially enhanced, which improves detection robustness in complex agricultural scenarios.

Driven by the advantages of multimodal fusion, an increasing number of studies have integrated visible and thermal information for UAV-based TL inspection. To enhance the perception capability of models, recent studies have explored various optimization strategies. For instance, Cheng et al. introduced the PL-UNeXt [6], which incorporates an edge detail head and a dynamic line feature head to guide the learning of structural shape priors. Wang et al. [7] designed a hybrid encoding architecture that combines dynamic snake convolution with global modeling mechanisms. Furthermore, Zhang et al. [8] and Guo et al. [9] proposed lightweight frameworks and knowledge distillation strategies to improve deployment feasibility. Although these prior studies have achieved encouraging progress, most existing methods still formulate TLD as a generic semantic segmentation task based on conventional encoder–decoder architectures. Owing to the extremely high aspect-ratio characteristics of TLs, which continuously extend across large spatial regions while occupying only a few pixels in width, the structural information of these lines is highly susceptible to degradation during the progressive downsampling process of standard networks. Moreover, under the severe inter-modal interference caused by complex crop textures and dense vegetation, these approaches remain insufficient for effectively modeling long-range contextual dependencies.

To address the structural degradation and semantic ambiguity limitations of existing RGB-TIR TLD methods, we propose AgriSafeNet, a multi-branch collaborative framework specifically designed for smart agricultural environments. Unlike conventional approaches based on standard encoder–decoder architectures and hierarchical feature pyramids, AgriSafeNet fundamentally redesigns the CNN–Transformer hybrid paradigm from the perspective of ultrathin target preservation, thereby simultaneously emphasizing local geometric continuity and global contextual modeling during multimodal feature extraction. Specifically, a customized dual-branch architecture is constructed for geometric structure preservation. In the Detail (DTA) branch, all downsampling operations are removed to retain high-resolution feature representations, thereby effectively preserving the spatial continuity and edge integrity of fine-grained TL structures. In contrast, the Context (CTX) branch employs the architectural design of SegFormer as the backbone to capture robust global contextual information by modeling long-range dependencies and complex semantic relationships across large spatial regions through self-attention mechanisms. By integrating fine-grained local structural features with global multimodal semantic representations, AgriSafeNet establishes more robust semantic associations between TLs and the surrounding agricultural environment, significantly enhancing the capability of the network to distinguish actual TLs from structurally similar background interference. Benefiting from this local–global collaborative representation mechanism, the proposed framework effectively improves segmentation completeness, contextual consistency, and overall detection robustness in complex agricultural scenarios without compromising fine-grained geometric details, thus further ensuring the operational safety of UAVs under adverse environmental conditions.

To further enhance multimodal representation learning and contextual reasoning capabilities for TLD, three dedicated modules are introduced: the multiscale feature decoupling and fusion module (MFDFM), the spatial compression attention fusion module (SCAFM), and the context-guided feature extraction module (CGFEM). Each module is specifically tailored to address the unique geometric and environmental challenges inherent in the extraction of ultrathin TLs.

The MFDFM is designed to overcome a common limitation of existing multimodal fusion methods, which frequently fail to explicitly distinguish between modality-shared information and modality-specific complementary features. Conventional RGB-TIR fusion approaches typically rely on direct feature aggregation, which tends to introduce redundant background noise and suppress critical discriminative cues of slender targets. To address this issue, multi-scale atrous convolution blocks (MACBs) are utilized to capture multi-scale responses, and a novel feature decoupling and fusion block (FDFB) is proposed. Before executing adaptive cross-modal fusion, the FDFB explicitly disentangles shared semantic information from modality-specific features. By isolating the thermal radiation characteristics of TLs from visual background clutter, this decoupled

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representation learning significantly enhances the interaction between complementary features, thereby improving the overall discriminability of the network. Furthermore, a cross-modal attention (CMA) mechanism based on a Non-Local structure [10] is incorporated to comprehensively capture multimodal contextual dependencies along the extensive length of TLs.

To further enhance deep semantic interaction and global contextual perception, the SCAFM is proposed to address the limited contextual modeling capabilities of conventional attention-based fusion methods. Unlike traditional spatial or channel attention mechanisms that primarily focus on local feature enhancement, the SCAFM utilizes horizontal spatial compression attention (HSA) and vertical spatial compression attention (VSA) [11]. These operations are specifically adapted to explicitly capture the extreme aspect ratios and elongated geometric priors of TLs. By combining average pooling (AvP) and max pooling (MaP), the module simultaneously captures global semantic distributions and fine-grained local structural responses. This design enables the SCAFM to perform context-aware multimodal fusion while effectively preserving the structural continuity that is essential for the segmentation of ultrathin targets across complex agricultural backgrounds.

Moreover, to improve feature robustness and environmental adaptability, the CGFEM is integrated into the AgriSafeNet framework. Unlike conventional feature enhancement methods that primarily focus on local response refinement, the CGFEM adaptively highlights semantic regions related to TLs while suppressing structurally confusing background noise. Consequently, visual interference caused by vegetation occlusions, cluttered textures, and field boundaries is effectively mitigated. By leveraging this dynamic context-guided representation, the proposed framework achieves significant improvements in segmentation completeness, contextual consistency, and overall robustness, particularly in challenging agricultural environments characterized by severe structural ambiguity. Finally, the refined multimodal representations are processed by the decoder to generate high-quality segmentation results for accurate obstacle avoidance.

Beyond algorithmic limitations, the scarcity of high-quality annotated datasets remains a substantial bottleneck for multimodal TLD research. To address this data scarcity and comprehensively evaluate the proposed framework, a novel RGB-TIR dataset, termed AgriSafe-TLD, is constructed for agricultural TLD tasks. The dataset was acquired utilizing a DJI Mavic 3T dual-spectrum UAV platform. AgriSafe-TLD provides high-quality multimodal imagery and serves as a critical foundation for the training and evaluation of TLD algorithms in this study, thereby promoting the further development of UAV-based intelligent inspection technologies for smart agricultural applications.

In summary, the primary contributions of this study are summarized as follows:

1) We propose AgriSafeNet, a novel RGB-TIR TLD architecture tailored for smart agricultural applications, is proposed. By synergistically integrating fine-grained structural features with global multimodal contextual information, this framework significantly enhances detection accuracy and structural completeness within complex farmland environments.
2) We develop the MFDFM to facilitate efficient shallow-level cross-modal interactions, and introduce the SCAFM to refine the integration of deep semantic features. This hierarchical fusion strategy enables robust cross-modal perception and contextual reasoning across diverse scales.
3) We introduce the CGFEM to explicitly model the intricate semantic dependencies between TLs and the surrounding environment. This mechanism effectively suppresses background interference and significantly improves the discriminability and robustness of target representations.

The remainder of this article is organized as follows. Section II reviews the related work on RGB-based and multimodal TLD. Section III details the overall architecture and key components of the proposed AgriSafeNet framework. Section IV presents the experimental results accompanied by a comprehensive performance analysis. Finally, Section V concludes this article and outlines potential directions for future research.

II. Related Work

A. RGB Image TLD

With the rapid advancement of RGB image processing and artificial intelligence, deep learning models have been increasingly adopted in electric power systems due to their strong capabilities in representation learning and feature extraction. These developments have driven substantial progress in TLD research.

Zhou et al. [12] developed the first fully automated real-time UAV-based detection system for TLs, demonstrating the feasibility of UAV-assisted TLD. Mao et al. [13] proposed a semantic segmentation framework for TLD using binocular vision and feature pyramid networks, demonstrating that binocular vision provides richer visual cues compared to monocular imagery. Yetgin et al. [14] introduced recognition algorithms for TLs based on aerial images, which determine the presence of TLs to support aerial scene understanding. Yang et al. [15] designed a U-shaped network with a VGG-16 backbone to extract TL features and integrated multiscale information during the decoding process to generate final predictions. Although skip connections are employed to partially recover spatial resolution, critical structural information is still lost during the downsampling process. Han et al. [16] proposed a lightweight TLD model based on GhostNet and utilized a shuffle attention mechanism to improve detection accuracy. Xu et al. [17] developed a multilevel feature aggregation module combined with attention mechanisms to detect TLs with varying widths and orientations, enhancing semantic representation while suppressing background noise. They further incorporated a spatial pyramid pooling module with an enlarged receptive field to fuse multiscale features for the final prediction. Gao et al. [18] introduced a multi-branch lightweight network consisting of a contextual branch for capturing global information and a spatial branch for preserving

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high-resolution details, achieving accurate and efficient TLD. Pan et al. [19] proposed a CNN-based classifier to eliminate background interference and identify image patches containing TLs, followed by a Hough transform-based refinement module for precise localization. Li et al. [20] developed an attention-based convolutional network with an encoder–decoder fusion architecture, effectively leveraging both semantic and positional information to focus on TLs rather than background regions. However, this method neglects the complementary contextual cues provided by the surrounding background, which are crucial for generating complete and continuous detection results.

To alleviate the high cost of pixel-level annotation, Lee et al. [21] proposed a weakly supervised learning framework for TL localization using only image-level labels. To address the challenges posed by the slender geometry of TLs and the complexity of surrounding environments, An et al. [22] introduced DUFormer, which improves feature extraction, enhancement, and binary segmentation through a relabeling encoder, a perception block, and an enhanced SCSE module. Although these approaches have achieved promising performance in single-modal TLD, the robustness of these methods remains limited under adverse conditions. With recent advances in sensing technology, RGB-TIR multimodal TLD models have therefore attracted increasing research attention.

B. RGB-TIR Image TLD

RGB-TIR TLD models effectively exploit the complementary information provided by visible and thermal infrared modalities to mitigate semantic ambiguities in RGB imagery under challenging illumination conditions, thereby achieving robust detection performance, particularly in complex environments.

Specifically, Choi et al. [23] enhanced the Fast-SCNN architecture by introducing additional convolutional and transposed convolutional layers to develop a deep learning-based real-time TLD network. They also constructed the first paired UAV-based visible and thermal infrared TLD dataset. To further improve the efficiency of cross-modal fusion, Choi et al. [24] proposed an attention-based multimodal feature fusion approach comprising a multi-branch feature extraction module and a channel attention module. The multi-branch feature extraction module captures representative features from each modality through parallel branches, which are subsequently aggregated into attention vectors within the channel attention module to adaptively recalibrate the input features. Xu et al. [25] introduced a dual-space graph interaction network for TLD, which employs two independent backbone networks to extract modality-specific features. High-level multimodal features are fused in the coordinate space, while feature clustering and allocation are conducted in the demand space. By leveraging both cross-graph and intragraph interactions, this model captures rich contextual dependencies for feature enhancement and inference. In addition, multitask learning is incorporated to address scale variations in thermal imagery and further improve detection accuracy. Beyond specific TLD applications, the synergistic fusion of multimodal remote sensing imagery has recently catalyzed significant breakthroughs across broader earth observation and smart agriculture tasks. For instance, in the agricultural domain, RustQNet [26] leverages a mutual information minimization module to compactly fuse RGB, multispectral, and vegetation index data, enabling the high-throughput quantitative assessment of crop diseases. In large-scale remote sensing classification, CCR-Net [27] introduces an advanced cross-channel reconstruction strategy that facilitates effective information exchange across modalities, such as hyperspectral and LiDAR/SAR data, to learn more compact and discriminative representations. Furthermore, the evolution of remote sensing foundation models, such as FlexiMo [28], has demonstrated exceptional flexibility by employing spatial resolution-aware and lightweight channel adaptation modules to dynamically process heterogeneous multimodal and multiscale imagery. These advanced paradigms underscore the immense potential of deeply decoupled and dynamically fused multimodal architectures.

Despite these advances in both general remote sensing and specific TLD applications, most existing approaches still formulate RGB-TIR TLD as a conventional semantic segmentation or saliency detection problem, thereby overlooking the distinctive slender and elongated structural characteristics of TLs. To overcome this limitation, we propose AgriSafeNet, a novel multimodal TLD framework that adopts a multi-branch architecture to jointly integrate fine-grained local features and global contextual information. This design strengthens the semantic association between TLs and the surrounding regions, effectively suppresses irrelevant background noise, and preserves critical structural details. Consequently, AgriSafeNet reduces false detections and background interference, leading to more complete and accurate prediction results.

III. Proposed Network

A. Unimodal Feature Extraction

AgriSafeNet employs a CNN-based encoder in the DTA branch to extract fine-grained local features, while a Transformer-based encoder is adopted in the CTX branch to capture and integrate contextual information.

The DTA branch is specifically designed without additional downsampling operations to preserve fine-grained local details. As illustrated in Fig. 2, the stem block consists of two "Conv-BN-ReLU" sequences, each reducing the spatial resolution by half. Each DTA block is implemented as a CNN module with a residual structure. Except for the stem block, no further downsampling is introduced in the DTA branch, thereby maintaining the feature resolution at one-quarter of the original input resolution and preserving critical spatial and geometric information. Consequently, four hierarchical levels of local features are extracted for both the RGB and TIR modalities, which are denoted as $\{\mathbf{F}_{dta}^{rgb,i} | i = 0, 1, 2, 3\}$ and $\{\mathbf{F}_{dta}^{tir,i} | i = 0, 1, 2, 3\}$, respectively.

To effectively capture rich contextual information, the CTX branch adopts the architectural design of SegFormer [29] as its backbone. Specifically, the output feature map from the first DTA block is fed into the CTX branch as input. The spatial

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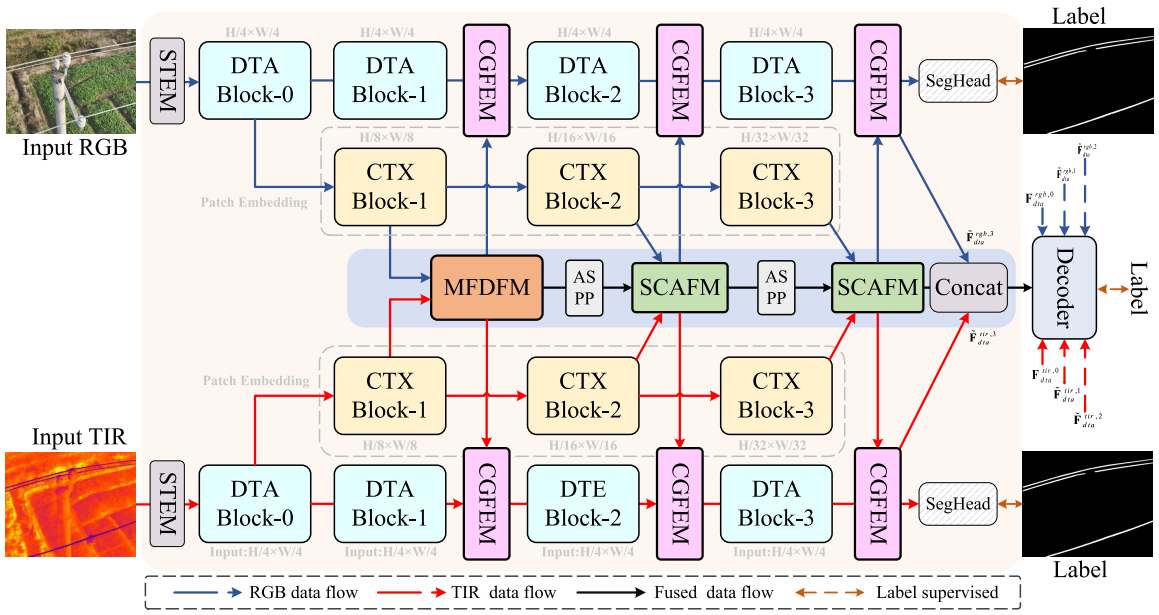


Fig. 2. Overall architecture of the proposed AgriSafeNet. The framework adopts a multi-branch network design that integrates convolutional neural networks and transformers as unimodal feature extractors to capture hierarchical representations from RGB and TIR inputs. Decoder-level supervision is omitted for clarity. MFDFM denotes the multi-scale feature decoupling and fusion module, SCAFM represents the spatial compression attention fusion module, and CGFEM refers to the context-guided feature extraction module.

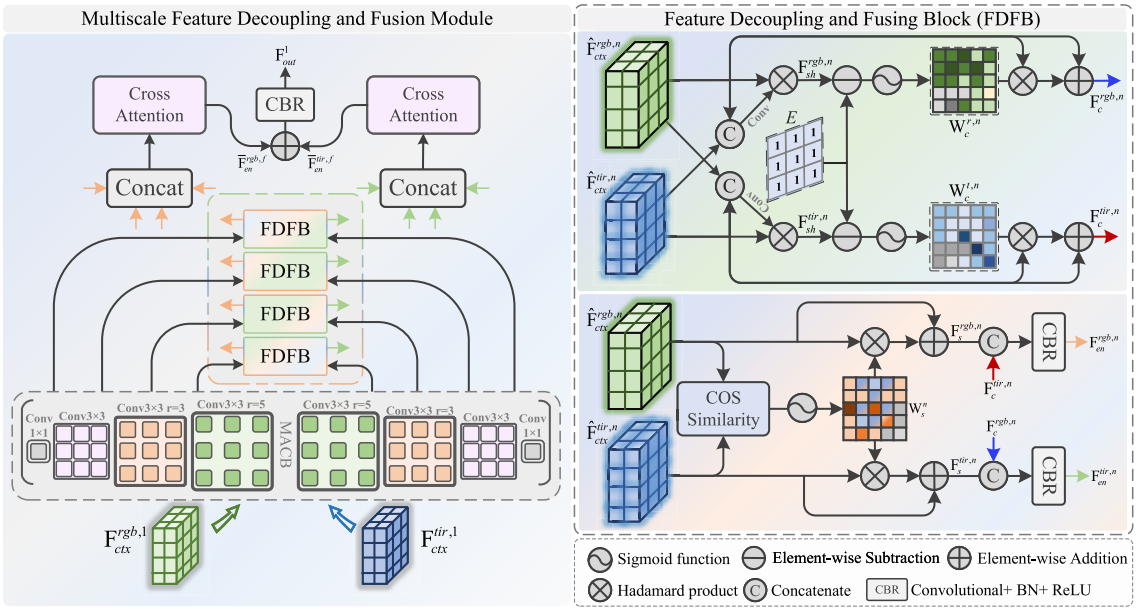


Fig. 3. Structure of the proposed MFDFM and the FDFB.

dimensions of the resulting feature maps are indicated in gray in Fig. 2, while the channel dimensions remain consistent with the baseline. After passing through a sequence of Transformer blocks, three levels of contextual features are obtained for the RGB and TIR modalities, denoted as $\{\mathbf{F}_{ctx}^{rgb,i}|i = 1,2,3\}$ and $\{\mathbf{F}_{ctx}^{tir,i}|i = 1,2,3\}$.

B. Contextual Feature Fusion

As discussed in Section I, rich contextual information plays a critical role in accurately distinguishing the semantic characteristics of TLs. To effectively integrate contextual cues from different modalities and reduce pixel-level classification errors in the detection of TLs, two complementary fusion strategies are introduced for shallow-level and deep-level contextual information.

1) Multiscale Feature Decoupling and Fusion Module: As illustrated in Fig. 3, to facilitate the effective reasoning and fusion of multimodal contextual information, we propose the MFDFM. This module exploits dilated convolutional pyramids to extract rich contextual cues across multiple scales and employs an FDFB to enhance the discriminability of features at different receptive fields. Furthermore, CMA interactions are introduced to comprehensively capture and integrate multimodal contextual information.

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Specifically, the CTX branch features $\mathbf{F}_{ctx}^{1,rgb}$ and $\mathbf{F}_{ctx}^{1,tir}$ are first fed into two MACBs with shared parameters to extract multiscale single-modal contextual features, namely, $\{\hat{\mathbf{F}}_{ctx}^{rgb,n}|n = 0,1,2,3\}$ and $\{\hat{\mathbf{F}}_{ctx}^{tir,n}|n = 0,1,2,3\}$. This design maps features from different modalities but at the same scale into a unified feature space. Each MACB consists of four parallel branches, including one 1×1 convolutional branch and three dilated convolutional branches with dilation rates $\{1,3,5\}$. Mathematically, this process can be formulated as follows:

$$\hat{\mathbf{F}}_{ctx}^{rgb,0}, \hat{\mathbf{F}}_{ctx}^{rgb,1}, \hat{\mathbf{F}}_{ctx}^{rgb,2}, \hat{\mathbf{F}}_{ctx}^{rgb,3} = \text{MACB}\left(\mathbf{F}_{ctx}^{rgb,1}\right) \tag{1}$$

$$\hat{\mathbf{F}}_{ctx}^{tir,0}, \hat{\mathbf{F}}_{ctx}^{tir,1}, \hat{\mathbf{F}}_{ctx}^{tir,2}, \hat{\mathbf{F}}_{ctx}^{tir,3} = \text{MACB}\left(\mathbf{F}_{ctx}^{tir,1}\right). \tag{2}$$

Directly employing simple fusion strategies to integrate features from different modalities may introduce feature redundancy and cause information loss, thereby restricting the effective exploitation of complementary cues across modalities. To overcome this limitation, the FDFB is designed to explicitly model and leverage both shared and complementary information between the modalities, which in turn enhances the discriminability of the features. Specifically, as illustrated in Fig. 3, the proposed FDFB is composed of two successive stages.

In the first stage, taking the RGB branch as an example, the RGB contextual features are first concatenated with the corresponding TIR contextual features to compute the shared representations, denoted as $\{\mathbf{F}_{sh}^{rgb,n}|n = 0,1,2,3\}$

$$\mathbf{F}_{sh}^{rgb,n} = \text{Conv}_{3\times3}\left(\text{Cat}\left(\hat{\mathbf{F}}_{ctx}^{rgb,n}; \hat{\mathbf{F}}_{ctx}^{tir,n}\right)\right) \otimes \hat{\mathbf{F}}_{ctx}^{rgb,n} \tag{3}$$

where $\text{Conv}_{3\times3}(*)$ denotes a convolutional block equipped with a 3×3 convolutional layer, which is used to adjust the number of feature channels. $\text{Cat}(*;*)$ represents the concatenation operation, and the symbol $\otimes$ denotes the Hadamard product. Next, the shared features $\mathbf{F}_{sh}^{rgb,n}$ are subtracted from an all-ones matrix to characterize the discrepancy information at the corresponding spatial locations, thereby yielding the complementary information weights $\{\mathbf{W}_c^{r,n}|n = 0,1,2,3\}$

$$\mathbf{W}_c^{r,n} = \text{Softmax}\left(\mathbf{E} - \mathbf{F}_{sh}^{rgb,n}\right) \tag{4}$$

where $\text{Softmax}(*)$ denotes the Softmax function, and $\mathbf{E}$ represents an all-ones matrix. In $\mathbf{W}_c^{r,n}$, larger values indicate stronger contextual complementarity between corresponding regions across different modalities, whereas smaller values reflect weaker complementarity. Based on the complementary information weights $\mathbf{W}_c^{t,n}$ and $\mathbf{W}_c^{r,n}$, the modality-specific complementary features for the TIR modality, denoted as $\{\mathbf{F}_c^{tir,n}|n = 0,1,2,3\}$, and for the RGB modality, denoted as $\{\mathbf{F}_c^{rgb,n}|n = 0,1,2,3\}$, are computed as follows:

$$\mathbf{F}_c^{rgb,n} = \mathbf{W}_c^{r,n} \otimes \hat{\mathbf{F}}_{ctx}^{rgb,n} + \hat{\mathbf{F}}_{ctx}^{rgb,n} \tag{5}$$

$$\mathbf{F}_c^{tir,n} = \mathbf{W}_c^{t,n} \otimes \hat{\mathbf{F}}_{ctx}^{tir,n} + \hat{\mathbf{F}}_{ctx}^{tir,n}. \tag{6}$$

In the second stage, the spatial similarity at the corresponding spatial locations is first quantified by computing the cosine similarity between $\{\hat{\mathbf{F}}_{ctx}^{rgb,n}|n = 0,1,2,3\}$ and $\{\hat{\mathbf{F}}_{ctx}^{tir,n}|n = 0,1,2,3\}$. Based on this operation, the corresponding contextual similarity weights $\{\mathbf{W}_s^n|n = 0,1,2,3\}$ are obtained as follows:

$$\mathbf{W}_s^n = \text{Softmax}\left(\text{Cos}\left(\hat{\mathbf{F}}_{ctx}^{rgb,n}; \hat{\mathbf{F}}_{ctx}^{tir,n}\right)\right) \tag{7}$$

where $\text{Cos}(*)$ denotes the cosine similarity operation. In $\mathbf{W}_s^n$, higher values indicate stronger contextual similarity between the corresponding regions across different modalities, while lower values imply weaker similarity. After obtaining the contextual similarity weights $\mathbf{W}_s^n$, the cross-modal contextual shared features $\{\mathbf{F}_s^{rgb,n}|n = 0,1,2,3\}$ and $\{\mathbf{F}_s^{tir,n}|n = 0,1,2,3\}$ can be computed as follows:

$$\mathbf{F}_s^{rgb,n} = \mathbf{W}_s^n \otimes \hat{\mathbf{F}}_{ctx}^{rgb,n} + \hat{\mathbf{F}}_{ctx}^{rgb,n} \tag{8}$$

$$\mathbf{F}_s^{tir,n} = \mathbf{W}_s^n \otimes \hat{\mathbf{F}}_{ctx}^{tir,n} + \hat{\mathbf{F}}_{ctx}^{tir,n}. \tag{9}$$

The contextual complementary features are then fused with the contextual shared features to generate the enhanced contextual representations, namely $\{\mathbf{F}_{en}^{rgb,n}|n = 0,1,2,3\}$ and $\{\mathbf{F}_{en}^{tir,n}|n = 0,1,2,3\}$

$$\mathbf{F}_{en}^{rgb,n} = \text{CBR}_{3\times3}\left(\text{Cat}\left(\mathbf{F}_s^{rgb,n}; \mathbf{F}_c^{rgb,n}\right)\right) \tag{10}$$

$$\mathbf{F}_{en}^{tir,n} = \text{CBR}_{3\times3}\left(\text{Cat}\left(\mathbf{F}_s^{tir,n}; \mathbf{F}_c^{tir,n}\right)\right) \tag{11}$$

where $\text{CBR}_{3\times3}(*)$ denotes a 3×3 convolutional layer followed by a batch normalization layer and a ReLU activation function. After obtaining the enhanced contextual features at four scales, namely, $\{\mathbf{F}_{en}^{rgb,n}|n = 0,1,2,3\}$ and $\{\mathbf{F}_{en}^{tir,n}|n = 0,1,2,3\}$, these features are first concatenated along the channel dimension. Thereafter, a 1×1 convolution is applied to fuse the multiscale contextual information. This process can be mathematically formulated as follows:

$$\mathbf{F}_{en}^{rgb,f} = \text{CBR}_{1\times1}\left(\text{Cat}\left(\mathbf{F}_{en}^{rgb,0}; \mathbf{F}_{en}^{rgb,1}; \mathbf{F}_{en}^{rgb,2}; \mathbf{F}_{en}^{rgb,3}\right)\right) \tag{12}$$

$$\mathbf{F}_{en}^{tir,f} = \text{CBR}_{1\times1}\left(\text{Cat}\left(\mathbf{F}_{en}^{tir,0}; \mathbf{F}_{en}^{tir,1}; \mathbf{F}_{en}^{tir,2}; \mathbf{F}_{en}^{tir,3}\right)\right). \tag{13}$$

Finally, a CMA operation is constructed within the MFDFM to enable the effective interaction and integration of information across different modalities. Specifically, the input feature maps with dimensions of $\mathbb{R}^{H\times W\times C}$ are first flattened into $\mathbb{R}^{N\times C}$, where $N = H \times W$. The flattened features are then linearly projected to generate the query (Q), key (K), and value (V) vectors.

A global attention interaction is then conducted between the K and V vectors of the current modality and the Q vector of the other modality, enabling the modeling of long-range dependencies among contextual features across different modalities. Mathematically, this process can be expressed as follows:

$$\mathbf{M}^{rgb} = \text{Softmax}\left(\text{P}\left(\mathbf{F}_{en}^{tir,f}; \alpha_q^t\right) \circ \text{P}\left(\mathbf{F}_{en}^{rgb,f}; \alpha_k^r\right)^T\right) \tag{14}$$

$$\mathbf{M}^{tir} = \text{Softmax}\left(\text{P}\left(\mathbf{F}_{en}^{rgb,f}; \alpha_q^r\right) \circ \text{P}\left(\mathbf{F}_{en}^{tir,f}; \alpha_k^t\right)^T\right) \tag{15}$$

$$\bar{\mathbf{F}}_{en}^{rgb,f} = \text{Reshape}\left(\mathbf{M}^{rgb} \circ \left(\text{P}\left(\mathbf{F}_{en}^{rgb,f}; \alpha_v^r\right)\right)\right) + \mathbf{F}_{en}^{rgb,f} \tag{16}$$

$$\bar{\mathbf{F}}_{en}^{tir,f} = \text{Reshape}\left(\mathbf{M}^{tir} \circ \left(\text{P}\left(\mathbf{F}_{en}^{tir,f}; \alpha_v^t\right)\right)\right) + \mathbf{F}_{en}^{tir,f} \tag{17}$$

where the symbol $\circ$ denotes the matrix multiplication, and $(*)^T$ represents the transpose of the matrix. $\text{P}(*;\alpha_q)$, $\text{P}(*;\alpha_k)$, and $\text{P}(*;\alpha_v)$ denote three linear projection layers parameterized by α_q, α_k, and α_v, respectively. Afterward, $\bar{\mathbf{F}}_{en}^{rgb,f}$ and $\bar{\mathbf{F}}_{en}^{tir,f}$ are separately fed into the CGFEM to enhance the spatial

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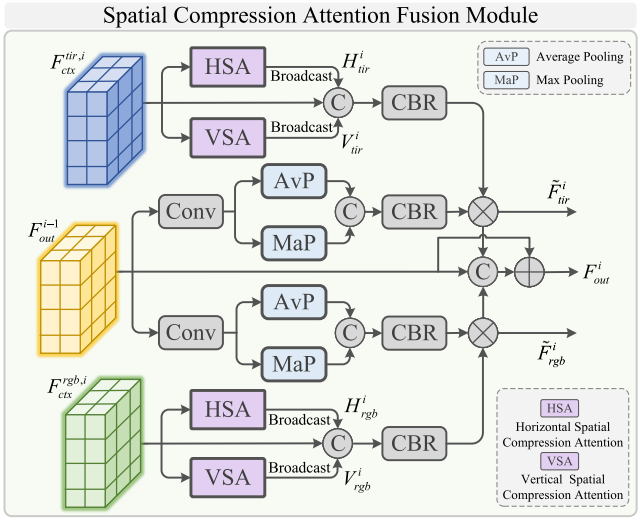


Fig. 4. Structure of the proposed SCAFM.

consistency and the discriminability of TL features within the DTA branch.

Meanwhile, the features $\bar{\mathbf{F}}_{en}^{rgb,f}$ and $\bar{\mathbf{F}}_{en}^{tir,f}$ are summed to generate $\mathbf{F}_{out}^{1}$

$$\mathbf{F}_{out}^{1} = \mathrm{CBR}_{3\times3}\left(\left(\bar{\mathbf{F}}_{en}^{rgb,f} + \bar{\mathbf{F}}_{en}^{tir,f}\right)\right) \tag{18}$$

which is subsequently fed into the first SCAFM for the inference and fusion of contextual information.

2) Spatial Compression Attention Fusion Module: To facilitate long-range contextual interactions across different modalities and to effectively integrate multilevel information, a novel SCAFM is proposed. The overall architecture of the SCAFM is illustrated in Fig. 4. The inputs to the SCAFM consist of the contextual features from the CTX branch, namely, $\{\mathbf{F}_{ctx}^{rgb,i}|i = 2,3\}$ and $\{\mathbf{F}_{ctx}^{tir,i}|i = 2,3\}$, together with the output features $\{\mathbf{F}_{out}^{i-1}|i = 2,3\}$ generated by the preceding module. Specifically, the SCAFM first applies convolutional operations with non-shared parameters to project $\{\mathbf{F}_{out}^{i-1}|i = 2,3\}$ into distinct feature spaces. Following this, the AvP and MaP operations are performed on the projected features. AvP captures global contextual information, whereas MaP highlights locally salient responses. The joint use of these pooling operations enables the effective integration of global and local cues, suppresses irrelevant background responses, and enhances the discriminability of the contextual representations. The mathematical formulation is given as follows:

$$\begin{aligned}\dot{\mathbf{F}}_{1}^{i} = \mathrm{Conv}_{1\times1}\Big(\mathrm{Cat}\Big(&\mathrm{AvP}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_{out}^{i-1}\right)\right)\\ &\mathrm{MaP}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_{out}^{i-1}\right)\right)\Big)\Big)\end{aligned} \tag{19}$$

$$\begin{aligned}\dot{\mathbf{F}}_{2}^{i} = \mathrm{Conv}_{1\times1}\Big(\mathrm{Cat}\Big(&\mathrm{AvP}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_{out}^{i-1}\right)\right)\\ &\mathrm{MaP}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_{out}^{i-1}\right)\right)\Big)\Big)\end{aligned} \tag{20}$$

where AvP$(*)$ and MaP$(*)$ denote the operations of AvP and MaP, respectively.

Furthermore, inspired by [11], horizontal and vertical spatial compression attention mechanisms are introduced. These mechanisms effectively capture the correlations between different positions in the feature map, aiding the model in comprehensively understanding the global structure of the image and improving the spatial consistency of TLs. Taking the RGB branch as an example, the axis-enhanced contextual features $\{\mathbf{V}_r^i|i = 2,3\}$ and $\{\mathbf{H}_r^i|i = 2,3\}$ can be calculated as follows:

$$\begin{aligned}\mathbf{V}_r^i = \mathrm{Att}\Big(&\mathrm{mean}_v\left(\mathrm{Pro}_q^V\left(\mathbf{F}_{ctx}^{rgb,i};\alpha_q^V\right)\right)\\ &\mathrm{mean}_v\left(\mathrm{Pro}_k^V\left(\mathbf{F}_{ctx}^{rgb,i};\alpha_k^V\right)\right)\\ &\mathrm{mean}_v\left(\mathrm{Pro}_v^V\left(\mathbf{F}_{ctx}^{rgb,i};\alpha_v^V\right)\right)\Big)\end{aligned} \tag{21}$$

$$\begin{aligned}\mathbf{H}_r^i = \mathrm{Att}\Big(&\mathrm{mean}_h\left(\mathrm{Pro}_q^H\left(\mathbf{F}_{ctx}^{rgb,i};\alpha_q^H\right)\right)\\ &\mathrm{mean}_h\left(\mathrm{Pro}_k^H\left(\mathbf{F}_{ctx}^{rgb,i};\alpha_k^H\right)\right)\\ &\mathrm{mean}_h\left(\mathrm{Pro}_v^H\left(\mathbf{F}_{ctx}^{rgb,i};\alpha_v^H\right)\right)\Big)\end{aligned} \tag{22}$$

where mean_v and mean_h denote the mean operations performed along the vertical and horizontal directions, respectively.

Afterward, the axis-enhanced RGB and TIR features, namely, $\{\tilde{\mathbf{F}}_{rgb}^i|i = 2,3\}$ and $\{\tilde{\mathbf{F}}_{tir}^i|i = 2,3\}$, are obtained as follows:

$$\tilde{\mathbf{F}}_{rgb}^i = \mathrm{CBR}_{3\times3}\left(\mathrm{Cat}\left(\mathrm{B}\left(\mathbf{H}_r^i\right);\mathrm{B}\left(\mathbf{V}_r^i\right);\mathbf{F}_{ctx}^{rgb,i}\right)\right)\otimes\dot{\mathbf{F}}_1^i \tag{23}$$

$$\tilde{\mathbf{F}}_{tir}^i = \mathrm{CBR}_{3\times3}\left(\mathrm{Cat}\left(\mathrm{B}\left(\mathbf{H}_t^i\right);\mathrm{B}\left(\mathbf{V}_t^i\right);\mathbf{F}_{ctx}^{tir,i}\right)\right)\otimes\dot{\mathbf{F}}_2^i \tag{24}$$

where Att$(*)$ denotes the non-local attention operation, and B$(*)$ represents the tensor broadcasting operation used to align feature dimensions for the subsequent fusion. Building upon these operations, the axis-enhanced features $\{\tilde{\mathbf{F}}_{rgb}^i|i = 2,3\}$ and $\{\tilde{\mathbf{F}}_{tir}^i|i = 2,3\}$ interact with the contextual information propagated from the preceding layer, enabling the effective fusion of contextual cues across different levels and modalities. Consequently, the multimodal contextual features $\{\mathbf{F}_{out}^i|i = 2,3\}$ are computed as follows:

$$\mathbf{F}_{out}^i = \mathrm{Cat}\left(\tilde{\mathbf{F}}_{rgb}^i,\tilde{\mathbf{F}}_{tir}^i,\mathbf{F}_{out}^{i-1}\right) + \mathbf{F}_{out}^{i-1}. \tag{25}$$

Next, $\tilde{\mathbf{F}}_{rgb}^i$ and $\tilde{\mathbf{F}}_{tir}^i$ are fed into the CGFEM to enrich the contextual representation within the DTA branch. Simultaneously, the multimodal contextual features $\mathbf{F}_{out}^i$ are forwarded to the subsequent SCAFM to further enhance the cross-level contextual interactions.

C. Context-Guided Feature Extraction Module

As illustrated in Fig. 5, the CGFEM is integrated into the proposed AgriSafeNet architecture to robustly capture the topological features of TLs and facilitate the assimilation of multimodal contextual information.

The intrinsic morphological characteristics of TLs impose substantial challenges on feature extraction. Specifically, the slender and curvilinear geometry of TLs prevents traditional convolutional methods from capturing complete structural information, frequently leading to spatial discontinuities in the feature maps. Moreover, the fine-grained nature of TLs renders the representations of these lines highly susceptible to information loss during downsampling operations. To mitigate these issues, the CGFEM incorporates dynamic snake convolutions, which adaptively deform the receptive fields to align with the distinctive geometric properties of TLs. Simultaneously, the

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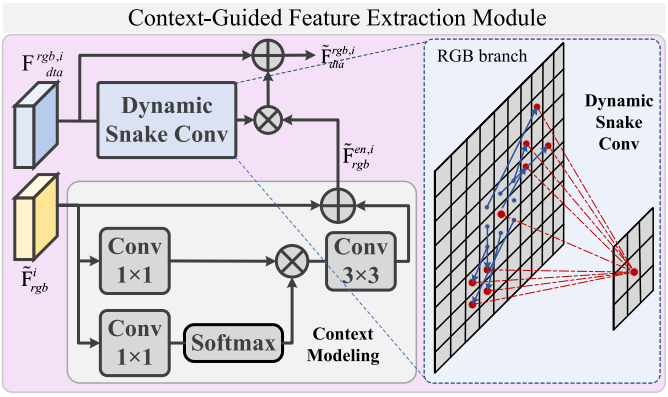


Fig. 5. Structure of the proposed CGFEM.

DTA branch maintains high-resolution feature representations to preserve the integrity of the fine-grained details of TLs.

However, relying solely on high-resolution inference restricts the effective receptive field, thereby limiting the capacity of the network to model long-range contextual dependencies. This lack of global context often results in the misclassification of TLs and confusion with visually similar background structures. To address this limitation, the CGFEM integrates multimodal contextual cues to explicitly model the semantic relationships between TLs and the surrounding environment. This mechanism augments the capability of the network to resolve complex scene interactions, significantly enhancing the overall detection performance.

The detailed structure of the CGFEM is illustrated in Fig. 5. Specifically, given the enhanced RGB and TIR features, i.e., $\{\tilde{\mathbf{F}}^{i}_{rgb}|i = 1,2,3\}$ and $\{\tilde{\mathbf{F}}^{i}_{tir}|i = 1,2,3\}$, as well as the local features at the current level, i.e., $\{\mathbf{F}^{rgb,i}_{dta}|i = 1,2,3\}$ and $\{\mathbf{F}^{tir,i}_{dta}|i = 1,2,3\}$, the CGFEM aims to effectively integrate fine-grained local details with global contextual information.

To explicate the contextual dependencies, a global attention map is initially derived to encode global contextual information via a weighted aggregation across all spatial positions. Following this, the aggregated global context is distributed to each spatial location through element-wise addition, thereby enriching the feature representation with global semantics. It is important to note that bilinear interpolation is employed to upsample the feature maps from the CTX branch to resolve the discrepancies in spatial resolution. For the sake of notational brevity, this upsampling operation is omitted in the subsequent formulation. Without loss of generality, taking the RGB branch as a representative example, this process can be formally expressed as follows:

$$\tilde{\mathbf{F}}^{en,i}_{rgb} = \mathrm{Conv}_{3\times3}\left(\mathrm{Softmax}\left(\mathrm{Reshape}\left(\mathrm{Conv}_{1\times1}\left(\tilde{\mathbf{F}}^{i}_{rgb}\right)\right)\right) \otimes \mathrm{Conv}_{1\times1}\left(\tilde{\mathbf{F}}^{i}_{rgb}\right)\right) + \tilde{\mathbf{F}}^{i}_{rgb}. \quad (26)$$

Subsequently, the dynamic snake convolution [30] is employed to adaptively extract the distinctive features of TLs. By interacting with the enhanced contextual features $\{\tilde{\mathbf{F}}^{en,i}_{rgb}|i = 1,2,3\}$, the contextual relationships between the representations of TLs and the cross-modal features are effectively established. Formally, the output RGB features $\{\tilde{\mathbf{F}}^{rgb,i}_{dta}|i = 1,2,3\}$ are computed as follows:

$$\tilde{\mathbf{F}}^{rgb,i}_{dta} = \mathrm{DSConv}\left(\mathbf{F}^{rgb,i}_{dta}\right) \otimes \tilde{\mathbf{F}}^{en,i}_{rgb} + \mathbf{F}^{rgb,i}_{dta} \quad (27)$$

where DSConv$(*)$ denotes the dynamic snake convolution operation.

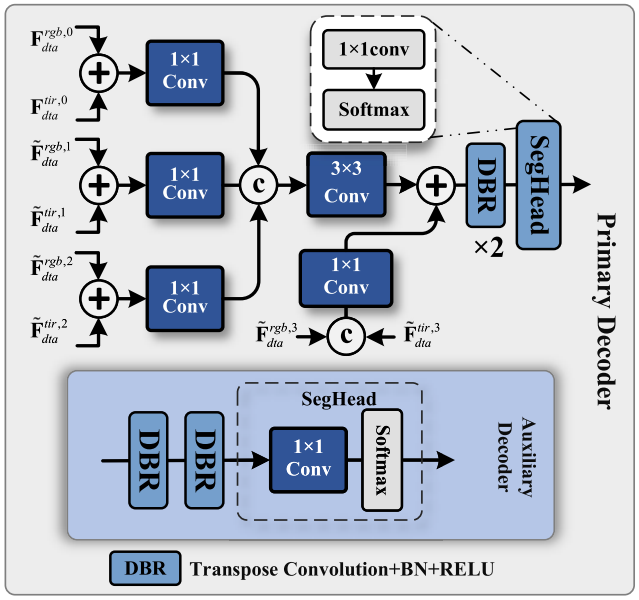


Fig. 6. Detailed architecture of the primary decoder and auxiliary decoders.

D. Decoder and Output Inference

The detailed architecture of the primary decoder and the auxiliary decoders is illustrated in Fig. 6. The outputs $\tilde{\mathbf{F}}^{rgb,3}_{dta}$ and $\tilde{\mathbf{F}}^{tir,3}_{dta}$ from the final CGFEM of the RGB and TIR branches are first concatenated and fused to obtain $\mathbf{F}^{f,3}_{dta}$. Subsequently, the output features $\mathbf{F}^{rgb,0}_{dta}$ and $\mathbf{F}^{tir,0}_{dta}$ from DTA-Block-0 of the respective modality branches are aggregated to obtain $\mathbf{F}^{f,0}_{dta}$, while the outputs from the CGFEMs at the first two levels are integrated to yield $\{\mathbf{F}^{f,i}_{dta}|i = 1,2\}$.

The aggregated features $\{\mathbf{F}^{f,i}_{dta}|i = 0,1,2\}$ are then utilized to restore the spatial information, facilitating the generation of the final primary prediction $\mathbf{S}_{pre}$. Mathematically, this process can be expressed as follows:

$$\mathbf{F}^{f,3}_{dta} = \mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(\tilde{\mathbf{F}}^{rgb,3}_{dta}, \tilde{\mathbf{F}}^{tir,3}_{dta}\right)\right) \quad (28)$$

$$\mathbf{F}^{f,0}_{dta} = \mathrm{Conv}_{1\times1}\left(\mathbf{F}^{rgb,0}_{dta} + \mathbf{F}^{tir,0}_{dta}\right) \quad (29)$$

$$\mathbf{F}^{f,1}_{dta} = \mathrm{Conv}_{1\times1}\left(\tilde{\mathbf{F}}^{rgb,1}_{dta} + \tilde{\mathbf{F}}^{tir,1}_{dta}\right) \quad (30)$$

$$\mathbf{F}^{f,2}_{dta} = \mathrm{Conv}_{1\times1}\left(\tilde{\mathbf{F}}^{rgb,2}_{dta} + \tilde{\mathbf{F}}^{tir,2}_{dta}\right) \quad (31)$$

$$\mathbf{F}_{final} = \mathrm{Conv}_{3\times3}\left(\mathrm{Cat}\left(\mathbf{F}^{f,0}_{dta}, \mathbf{F}^{f,1}_{dta}, \mathbf{F}^{f,2}_{dta}\right)\right) + \mathbf{F}^{f,3}_{dta} \quad (32)$$

$$\mathbf{S}_{pre} = \mathrm{SegHead}\left(\mathrm{DBR}\left(\mathrm{DBR}\left(\mathbf{F}_{final}\right)\right)\right) \quad (33)$$

where DBR$(*)$ denotes a 3×3 transposed convolutional layer followed by a batch normalization layer and a ReLU activation function, and SegHead$(*)$ denotes a 1×1 convolution, which is utilized to predict the primary probability map $\mathbf{S}_{pre}$.

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To ensure the reliability of feature extraction and to provide intermediate supervision, auxiliary decoders are applied to the deepest feature representations. Specifically, the output features $\tilde{\mathbf{F}}_{dta}^{rgb,3}$ and $\tilde{\mathbf{F}}_{dta}^{tir,3}$ are processed by independent SegHead branches to generate the auxiliary probability maps, denoted as $\mathbf{S}_{aux}^{rgb}$ and $\mathbf{S}_{aux}^{tir}$, respectively. This auxiliary prediction process is defined as follows:

$$\mathbf{S}_{aux}^{rgb} = \text{SegHead}\left(\text{DBR}\left(\text{DBR}\left(\tilde{\mathbf{F}}_{dta}^{rgb,3}\right)\right)\right) \tag{34}$$

$$\mathbf{S}_{aux}^{tir} = \text{SegHead}\left(\text{DBR}\left(\text{DBR}\left(\tilde{\mathbf{F}}_{dta}^{tir,3}\right)\right)\right). \tag{35}$$

These auxiliary predictions provide additional gradient flow during the training process, thereby enhancing the semantic discriminability of each unimodal branch without adding any computational burden during the testing phase.

E. Loss Function

The overall loss function $\mathcal{L}_{total}$ of AgriSafeNet comprises the primary loss $\mathcal{L}_p$ and two auxiliary losses, $\mathcal{L}_0$ and $\mathcal{L}_1$

$$\mathcal{L}_{total} = \mathcal{L}_p + \lambda_0\mathcal{L}_0 + \lambda_1\mathcal{L}_1 \tag{36}$$

where λ_0 and λ_1 are two hyperparameters utilized to balance the weights among the loss terms. Both λ_0 and λ_1 are empirically set to 0.3. Specifically, the final prediction map $\mathbf{S}_{pre}$ is supervised by the primary loss $\mathcal{L}_p$, while the auxiliary prediction maps ($\mathbf{S}_{aux}^{rgb}$ and $\mathbf{S}_{aux}^{tir}$) are supervised by $\mathcal{L}_0$ and $\mathcal{L}_1$. Due to the slender geometry of TLs, which occupy a minimal fraction of the total pixels, we employ a combination of the binary cross-entropy loss and the Dice loss as the loss function

$$\mathcal{L}_i = \alpha\mathcal{L}_{\text{BCE}} + (1-\alpha)\,\mathcal{L}_{\text{Dice}}, \quad i \in \{p, 0, 1\} \tag{37}$$

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log p_i + (1-y_i)\log(1-p_i)\right) \tag{38}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i p_i}{\sum_{i=1}^{N} y_i^2 + \sum_{i=1}^{N} p_i^2} \tag{39}$$

where α is a hyperparameter utilized to balance the weights between the binary cross-entropy loss and the Dice loss, which is empirically set to 0.5. In this context, N denotes the total number of pixels. The variable $y_i \in \{0, 1\}$ represents the ground-truth label of the i-th pixel, where 0 indicates the background and 1 indicates the foreground. Furthermore, $p_i \in [0, 1]$ denotes the predicted probability of the i-th pixel belonging to the foreground.

IV. Experiments

A. Evaluation Metrics

To comprehensively evaluate the detection performance of different methods, both region-based and topology-aware evaluation metrics are adopted in this work. Specifically, the conventional F-score and Intersection over Union (IoU) are employed to quantitatively assess the accuracy of pixel-level segmentation. The corresponding formulations are defined as follows:

$$\text{Precision} = \frac{TP_i}{TP_i + FP_i} \tag{40}$$

$$\text{Recall} = \frac{TP_i}{TP_i + FN_i} \tag{41}$$

$$\text{F-score} = \left(1+\beta^2\right) \times \frac{\text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \tag{42}$$

$$\text{IoU} = \frac{TP_i}{TP_i + FP_i + FN_i} \tag{43}$$

where β is set to 1, and TP_i, FP_i, and FN_i denote the numbers of true-positive, false-positive, and false-negative pixels for class i, respectively.

However, for extremely thin structures, such as TLs, the conventional pixel-wise metrics may not fully reflect structural continuity and localization reliability. In practical scenarios, slight spatial deviations between the predictions and the annotations may lead to substantial fluctuations in the strict pixel-level evaluation, even when the correct topological structures are preserved. To address this issue, we further introduce three tolerance-based evaluation metrics, namely, Correctness (Corr.), Completeness (Comp.), and Quality, which have been widely adopted in thin-object extraction tasks. By allowing a small spatial tolerance distance, the correctness and completeness metrics can be interpreted as relaxed precision and recall metrics, respectively. The formulations of these metrics are defined as follows:

$$\text{Corr.} = \frac{TP_{pred}}{N_{pred}} \tag{44}$$

$$\text{Comp.} = \frac{TP_{gt}}{N_{gt}} \tag{45}$$

$$\text{Quality} = \frac{\text{Comp.} \times \text{Corr.}}{\text{Comp.} - \text{Comp.} \times \text{Corr.} + \text{Corr.}} \tag{46}$$

where N_{pred} and N_{gt} denote the total numbers of predicted TL pixels and the ground-truth pixels, respectively. Furthermore, TP_{pred} represents the number of predicted pixels that fall within the tolerance neighborhood of the ground truth, whereas TP_{gt} denotes the number of ground-truth pixels located within the tolerance neighborhood of the predictions. Compared with the strict pixel-to-pixel evaluation, these tolerance-aware metrics provide a more reliable assessment of structural continuity and geometric localization accuracy for thin-object detection tasks.

B. Implementation

The proposed framework is implemented using the PyTorch library. All experiments are conducted on an NVIDIA GeForce RTX 4090 GPU equipped with 24 GB memory. To ensure stable convergence, the rectified Adam (RAdam) optimizer is utilized during the training process of the network. The learning rate is initialized to 5×10^{-5} and is dynamically adjusted via a polynomial decay policy throughout the 400 training epochs. Furthermore, to enhance generalization capability and mitigate overfitting, a comprehensive data augmentation strategy is employed. This strategy encompasses geometric transformations, such as random scaling, flipping, and cropping, as well as photometric distortions, including random adjustments to the brightness, contrast, and saturation of the images.

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TABLE I

QUANTITATIVE RESULTS (%) OF DIFFERENT MODELS ON THE AGRISAFE-TLD. THE BEST RESULTS ARE INDICATED IN BOLD. "–" REPRESENTS THAT THE RESULTS AND THE CORRESPONDING CODES ARE NOT OPEN-SOURCED

			Different weather conditions (IoU(%))				*Original images*			
Method	**Input**	**Venue**	**Day(%)**	**Night(%)**	**Fog(%)**	**Snow(%)**	**Pre(%)**	**Recall(%)**	**F1-Score(%)**	**IoU(%)**
Transmission Line Detection Method										
PL-UNeXt [6]	RGB	ICIP	40.90	38.71	39.95	40.33	66.42	53.22	59.09	41.94
PLGAN [31]	RGB	TIP	46.01	43.22	45.01	44.87	73.46	56.45	63.84	46.89
Real-time PL [23]	RGB+TIR	IVCNZ	44.72	42.48	43.73	43.91	79.24	51.55	62.46	45.42
UNet+FuseNet [32]	RGB+TIR	ACCV	43.54	41.67	42.88	42.36	79.40	50.41	61.67	44.58
UNet+MMTM [32]	RGB+TIR	CVPR	44.49	41.68	43.81	43.00	80.67	51.01	62.50	45.45
UNet+FFM [24]	RGB+TIR	TII	44.77	42.46	44.05	43.95	79.73	51.74	62.76	45.73
UMFNet+FuseNet [32]	RGB+TIR	ACCV	44.98	42.35	43.50	43.30	78.18	52.13	62.55	45.50
UMFNet+MMTM [32]	RGB+TIR	CVPR	45.52	42.83	44.61	44.56	80.22	52.56	63.51	46.53
UMFNet+FFM [24]	RGB+TIR	TII	46.62	44.56	45.90	45.22	80.01	54.00	64.48	47.58
RGB-TIR Semantic Segmentation Method										
MFNet [33]	RGB+TIR	IROS	44.37	41.29	43.31	42.87	70.35	54.93	61.69	44.61
RTFNet-152 [34]	RGB+TIR	RA-L	45.88	43.11	44.88	44.62	77.30	54.29	63.78	46.82
FDCNet [35]	RGB+TIR	TCSVT	47.52	44.84	46.88	46.65	74.98	57.92	65.35	48.54
FEANet [36]	RGB+TIR	IROS	45.55	42.96	44.51	44.64	79.22	52.57	63.20	46.20
GMNet [37]	RGB+TIR	TIP	45.67	43.06	44.45	43.96	81.01	52.11	63.42	46.44
SA-Gate [38]	RGB+TIR	ECCV	46.45	43.85	45.68	45.15	80.91	53.47	64.39	47.48
PSTNet [39]	RGB+TIR	ICRA	46.58	44.41	45.62	45.68	75.71	55.76	64.22	47.30
LASNet [40]	RGB+TIR	TCSVT	45.18	42.16	44.24	44.22	81.03	51.65	63.09	46.08
DHFNet [41]	RGB+TIR	IJON	44.01	41.83	43.02	42.82	**81.31**	50.14	62.03	44.96
GCNet [42]	RGB+TIR	IJON	45.50	43.48	44.71	44.58	81.25	52.17	63.54	46.57
FuseSeg [43]	RGB+TIR	TASE	47.77	45.43	46.60	46.78	77.11	56.67	65.33	48.51
ABMDRNet [44]	RGB+TIR	CVPR	47.90	44.96	46.80	47.04	80.66	55.03	65.42	48.61
SFAF-MA [45]	RGB+TIR	TIM	48.27	46.02	47.26	47.13	79.44	56.01	65.70	48.92
SCRNet [46]	RGB+TIR	PR	47.54	45.49	46.99	47.04	76.34	57.11	65.34	48.52
MCOFNet [47]	RGB+TIR	Evol. Syst.	47.77	45.32	46.03	46.19	74.71	57.25	64.82	47.95
AGFNet [48]	RGB+TIR	TITS	48.79	47.03	47.66	47.89	76.52	57.86	65.89	49.14
NTFNet [49]	RGB+TIR	APIN	48.27	46.31	47.34	46.94	75.31	58.82	66.05	49.31
AgriSafeNet(Ours)	RGB+TIR	-	**50.31**	**47.32**	**49.14**	**48.64**	72.42	**62.92**	**67.34**	**50.75**

C. Comparison With the State-of-the-Art Methods

To comprehensively evaluate the effectiveness and generalization capability of the proposed framework, extensive comparative experiments are conducted based on two benchmarks for RGB-TIR TLD, namely, the proposed AgriSafe-TLD dataset and the publicly available VITLD dataset. AgriSafe-TLD focuses on complex low-altitude agricultural environments with diverse illumination and weather conditions, whereas VITLD mainly targets high-altitude inspection scenarios for TLs. Detailed descriptions of both datasets are provided in the Supplementary Material.

1) AgriSafe-TLD: The proposed method is compared against a series of state-of-the-art approaches, including both dedicated methods for RGB-TIR TLD and representative models for RGB-TIR semantic segmentation. For a fair comparison, all competing semantic segmentation methods are adapted to the binary task of TLD by unifying the number of prediction categories to two classes, namely, TL and background. Unless otherwise specified, all methods are trained and evaluated under identical experimental settings.

To assess the generalization capability of AgriSafeNet under diverse weather conditions, multiple scenarios, including daytime, nighttime, fog, and snow, are evaluated during the testing phase using a combination of genuinely acquired images and randomly simulated data. All experimental results are obtained by independent runs to ensure the reliability and stability of the evaluation. It is important to note that while synthetic augmentations provide a highly representative proxy for complex weather, certain conclusions regarding model robustness are partially supported by these simulated perturbations, especially in the heavily simulated snowy scenarios. Therefore, we conservatively acknowledge that the actual robustness of the proposed framework under genuine extreme physical environments may necessitate further large-scale empirical validation in future studies. The overall quantitative results, together with the performance under each weather condition, are summarized in Table I. The results indicate that models specifically designed for RGB-TIR TLD consistently achieve superior performance compared with the approaches adapted from multimodal semantic segmentation methods. The relatively lower performance of the latter can be attributed to the corresponding training strategies and network architectures, which do not fully consider the unique morphological characteristics of TLs and the specific properties of thermal imaging data. The proposed AgriSafeNet achieves a substantial improvement and obtains the highest IoU among all compared state-of-the-art models.

Furthermore, standard pixel-wise evaluation metrics are often insufficient to fully reflect the topological continuity and structural integrity of extremely thin objects, such as TLs. The manual annotations of such wire-like structures inevitably contain minor pixel-level offsets, and slight spatial deviations in the predictions do not significantly compromise

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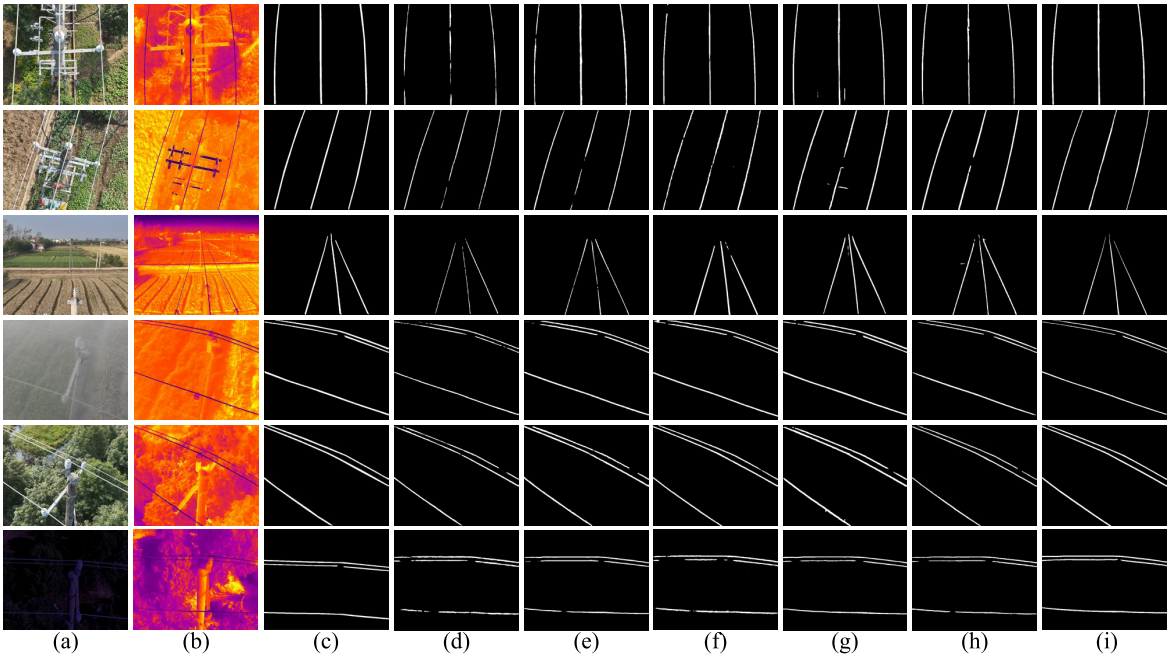


Fig. 7. Qualitative comparisons of different models on the AgriSafe-TLD. (a) RGB image. (b) TIR image. (c) GT. (d) MFNet. (e) RTFNet. (f) FEANet. (g) FDCNet. (h) MDRNet+. (i) Ours.

TABLE II
QUANTITATIVE COMPARISON USING SPECIALIZED THIN-OBJECT EVALUATION METRICS (CORR., COMP., AND QUALITY)

Method	Input	Corr.↑	Comp.↑	Quality ↑
MFNet	RGB+TIR	81.72	71.86	61.91
GCNet	RGB+TIR	82.81	71.65	62.37
RTFNet	RGB+TIR	85.97	70.62	63.32
SA-Gate	RGB+TIR	84.70	71.92	63.65
MCOFNet	RGB+TIR	85.44	72.19	64.28
PSTNet	RGB+TIR	84.63	72.81	64.31
FuseSeg	RGB+TIR	88.31	72.45	66.11
ABMDRNet	RGB+TIR	89.81	71.83	66.42
SFAF-MA	RGB+TIR	85.29	75.57	66.86
SCRNet	RGB+TIR	87.67	72.82	66.05
AGFNet	RGB+TIR	89.48	73.02	67.25
NTFNet	RGB+TIR	89.92	73.41	67.83
AgriSafeNet	RGB+TIR	89.39	75.61	**69.38**

practical safety. Therefore, to provide a more comprehensive and practically meaningful assessment, we additionally evaluate the models using specialized metrics for thin objects, namely, Correctness (Corr.), Completeness (Comp.), and Quality. Following the established evaluation protocols in previous studies on thin-object extraction, a 2-pixel tolerance mechanism is adopted, meaning that predictions falling within a 2-pixel neighborhood of the ground truth are regarded as valid. All tolerance-based evaluation results are obtained by independent experimental runs. As reported in Table II, AgriSafeNet demonstrates superior performance under these topology-aware metrics. While maintaining a highly competitive correctness score, the proposed method achieves new state-of-the-art results in completeness and overall quality, outperforming the second-best method by substantial margins. This significant improvement in completeness indicates that AgriSafeNet is highly effective in preserving the continuous structure of TLs and reducing fragmentation, which is critical for the safety of practical operations.

In addition, qualitative results produced by different models are presented in Fig. 7. These results demonstrate that AgriSafeNet generates more complete and accurate predictions of TLs under all tested weather conditions. This improvement can be attributed to the multi-branch network design. Within the CTX branch, the proposed MFDFM and SCAFM modules fully exploit cross-modal contextual complementarity. Within the DTA branch, the CGFEM further improves detection accuracy and structural completeness by integrating cross-modal contextual information into the inference process.

2) VITLD: Based on the VITLD dataset, the experimental results of the proposed AgriSafeNet are summarized in Table III. "ALL" in Table III represents the results obtained by testing on the original images combined with the randomly simulated images under daytime, nighttime, fog, and snow conditions. Notably, given the relatively uniform scenes in the VITLD dataset, the visible light images in the test set are dynamically augmented during the testing phase using the Python library *imgaug* to simulate diverse weather and illumination conditions, including daytime, nighttime, fog, and snow. To ensure the reliability and stability of the evaluation, all reported results are obtained by averaging multiple independent test runs. Under these simulated scenarios, AgriSafeNet consistently achieves superior performance, markedly surpassing all other compared methods. This performance advantage is primarily attributed to the carefully designed multi-branch network architecture, which effectively strengthens contextual interactions among pixels while preserving feature integrity, thereby serving as a critical factor for accurate pixel-level classification.

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TABLE III

MEAN IOU RESULTS (%) COMPARISON BASED ON VITLD. THE BEST RESULTS ARE INDICATED IN BOLD. "–" REPRESENTS THAT THE RESULTS ARE NOT OPEN-SOURCED. METHODS ABOVE THE MIDDLE DIVIDING LINE REPORT ORIGINAL BENCHMARK SCORES, WHEREAS METHODS BELOW PRESENT OUR REPRODUCED EVALUATIONS

Method	Day	Night	Fog	Snow	All
UNet+FuseNet	59.32	55.62	57.57	58.57	-
UNet+MMTM	59.61	57.38	58.08	59.20	-
UNet+FFM	60.89	58.00	59.00	60.59	-
UMF+FuseNet	62.05	60.47	60.79	61.53	-
UMF+MMTM	62.92	61.63	62.28	62.31	-
UMF+FFM	63.72	62.60	62.65	63.01	-
MFNet	39.38	38.22	38.17	39.14	38.85
FDCNet	47.72	43.23	41.33	41.07	44.21
ABMDRNet	52.24	50.13	51.02	50.55	51.23
MDRNet+	54.12	51.21	51.33	51.29	52.41
GMNet	56.97	48.77	54.22	53.74	54.13
GCNet	60.36	57.79	58.08	58.91	59.10
FuseSeg	61.32	54.45	58.21	58.40	58.74
DHFNet	62.22	57.23	60.01	61.93	60.72
SFAF-MA	58.04	57.49	58.01	57.99	57.91
NTFNet	63.14	58.01	60.33	62.12	60.91
AgriSafeNet	**66.12**	**63.05**	**64.72**	**65.56**	**65.78**

TABLE IV

QUANTITATIVE RESULTS OF ABLATION STUDY FOR EACH KEY COMPONENT IN THE PROPOSED MODEL. THE BEST RESULTS ARE INDICATED IN BOLD

Method	F1-Score(%)	IoU(%)
Baseline	62.33	45.27
B+CGFEM	64.71	47.83
B+MFDFM	63.93	46.98
B+SCAFM	63.81	46.85
B+MFDFM+SCAFM	64.79	47.92
B+CGFEM+SCAFM	65.17	48.33
B+CGFEM+MFDFM	66.41	49.71
B+CGFEM+MFDFM+SCAFM	**67.34**	**50.75**

D. Ablation Analyses

1) Effectiveness of Each Component: In this section, the comprehensive ablation experiments are conducted on the AgriSafe-TLD dataset to quantify the individual contributions of the key components within the proposed AgriSafeNet. To rigorously isolate the impact of each module and avoid an exclusively incremental evaluation, the ablation study is expanded to include all orthogonal combinations of the proposed modules, as summarized in Table IV.

Specifically, the CGFEM, MFDFM, and SCAFM modules are removed from the full AgriSafeNet architecture to construct a baseline model, denoted as "Baseline" (or "B"). In this baseline model, the cross-modal features are fused using element-wise addition. First, to assess the independent effectiveness of each proposed module, these modules are individually incorporated into the baseline model, resulting in the configurations "B+CGFEM," "B+MFDFM," and "B+SCAFM." A comparison between "B" and "B+CGFEM" demonstrates that embedding rich cross-modal contextual information into the feature extraction process substantially improves the performance of TLD and reduces pixel-level misclassifications.

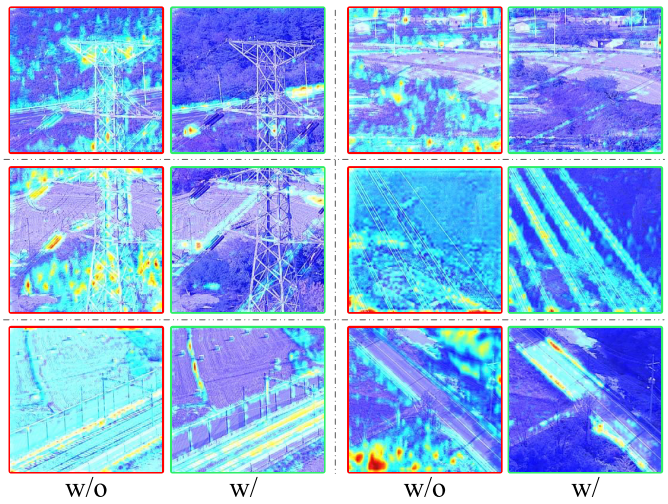


Fig. 8. Visual comparison of the features with and without CGFEM. For each set, the image within the red box on the left represents the feature visualization without CGFEM enhancement. The image within the green box of each set represents the feature visualization with CGFEM introduced.

Furthermore, visual analyses of the extracted features are conducted to intuitively illustrate the advantages of the CGFEM. As shown in Fig. 8, integrating the CGFEM into AgriSafeNet for the fusion of multimodal contextual information effectively suppresses irrelevant background noise, enabling the network to focus on the TLs and their immediate surroundings. Similarly, the evaluation of "B+MFDFM" clearly indicates that the proposed MFDFM substantially enhances the performance of RGB-TIR TLD. This improvement can be attributed to the capability of the MFDFM to strengthen the representation of features from different modalities by leveraging complementary information across four scales, thereby enabling a more comprehensive exploitation of shallow cross-modal contextual information. Additionally, the stand-alone inclusion of the SCAFM ("B+SCAFM") yields a marked improvement in TLD performance due to the module's capability to explicitly extract contextual information along multiple axes. Next, to evaluate the pairwise synergistic contributions of these components, two-module combinations are tested, namely, "B+CGFEM+MFDFM," "B+CGFEM+SCAFM," and "B+MFDFM+SCAFM." The quantitative results indicate that any combination of two modules outperforms the respective single-module configurations, demonstrating that the modules are mutually beneficial and extract complementary multimodal features. Finally, the complete architecture is assembled by integrating all three modules. The addition of the SCAFM to the "B+CGFEM+MFDFM" configuration enhances the modeling of deep contextual information through interaction with contextual features propagated from the previous network layer. Consequently, the discriminability and integrity of the multimodal contextual features at the current stage are significantly improved. Ultimately, the full AgriSafeNet model, represented as "B+CGFEM+MFDFM+SCAFM," achieves the best performance.

2) Effectiveness of Each Component of the MFDFM: To evaluate the effectiveness of each component within the MFDFM, comprehensive ablation experiments are conducted using four variant modules, as summarized in Table V. A

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TABLE V

QUANTITATIVE RESULTS (%) OF ABLATION STUDY FOR EACH KEY COMPONENT IN MFDFM. THE BEST RESULTS ARE INDICATED IN BOLD

Method	MACB	FDFB	CMA	F1-Score(%)	IoU(%)
MFDFM-V1	✓	✓	✓	**67.34**	**50.75**
MFDFM-V2	✓	✓		65.53	48.73
MFDFM-V3	✓		✓	65.87	49.11
MFDFM-V4	✓			65.49	48.69

check mark in the table indicates that the corresponding module is included in the MFDFM. The complete MFDFM is denoted as "MFDFM-V1." To assess the contribution of the CMA mechanism, this component is removed from the original MFDFM, and the contextual information from different modalities is fused through element-wise addition, resulting in the "MFDFM-V2" variant. To investigate the effectiveness of the FDFB, this module is removed from the original MFDFM, and the output of the MACB is directly cascaded and fed into the CMA mechanism, forming the "MFDFM-V3" variant. In the "MFDFM-V4" variant, both the FDFB and the CMA mechanism are removed.

A comparison of the results in Table V indicates that incorporating the CMA mechanism, as demonstrated by the comparison between "MFDFM-V1" and "MFDFM-V2," or "MFDFM-V3" and "MFDFM-V4," effectively models and fuses contextual information across modalities. Additionally, the performance difference between "MFDFM-V1" and "MFDFM-V3" highlights that the FDFB enhances the robustness of the feature representation at the current scale by exploiting complementary and modality-specific information. Moreover, the comparison between "MFDFM-V1" and "MFDFM-V4" indicates that the joint use of the FDFB and the CMA mechanism substantially improves the performance of TLD.

Overall, the experimental results of the four variants demonstrate that integrating the MFDFM into AgriSafeNet, employing the MACB for the extraction of multiscale contextual information, and leveraging both complementary and modality-specific features significantly enhance the discriminability of the fused multimodal context, thereby improving the performance of TLD.

3) Effectiveness of Each Component of the SCAFM: To comprehensively evaluate the effectiveness of each internal design component within the proposed SCAFM, several variant architectures are constructed, and quantitative experiments are conducted on the AgriSafe-TLD dataset, as summarized in Table VI. The complete architecture of the SCAFM consists of four key components, namely, HSA, VSA, AvP, and MaP. This complete configuration is denoted as "SCAFM-V1."

To investigate the contribution of the dual-axis spatial compression mechanism, both HSA and VSA are first removed while retaining AvP and MaP, resulting in the "SCAFM-V2" variant. To evaluate the effectiveness of the complementary pooling strategy, both AvP and MaP operations are removed from the complete model, forming the "SCAFM-V3" variant that relies solely on the spatial compression attention mechanism. Furthermore, to separately analyze the role of each spatial attention branch, two additional variants are constructed. Specifically, the "SCAFM-V4" variant removes the VSA branch while preserving HSA, AvP, and MaP, whereas the "SCAFM-V5" variant removes HSA while retaining VSA and the two pooling operations. As shown in Table VI, the complete "SCAFM-V1" configuration achieves the best quantitative performance across all the evaluation metrics, demonstrating the effectiveness of jointly integrating dual-axis spatial compression attention with complementary pooling operations. Compared with "SCAFM-V1," the removal of both HSA and VSA in "SCAFM-V2" causes the IoU to decrease from 50.75% to 49.79%, confirming that the spatial compression attention plays a crucial role in modeling the long-range contextual dependencies for the ultrathin structures of the TLs. Furthermore, the comparison between "SCAFM-V1" and "SCAFM-V3" demonstrates that the removal of the pooling operations results in a noticeable performance degradation. Specifically, the IoU decreases by 0.87%, indicating that the combination of the AvP and the MaP effectively captures both the global semantic distributions and the sparse local structural responses. Because the TLs typically occupy only a few pixels in the UAV imagery, relying solely on spatial attention without the complementary pooling operations may weaken the representation of the high-confidence local responses corresponding to the ultrathin structures. The comparisons among "SCAFM-V1," "SCAFM-V4," and "SCAFM-V5" further validate the necessity of the proposed dual-axis spatial compression design. The individual removal of either VSA or HSA leads to a noticeable performance degradation. These results indicate that the modeling of both the horizontal and vertical spatial dependencies is essential for maintaining the structural continuity of the TLs under varying spatial distributions and UAV viewing perspectives. In particular, the simultaneous modeling of the horizontal and vertical contextual relationships enables the network to better capture the long-range geometric continuity and suppress the structurally confusing background interference in complex agricultural environments.

TABLE VI

QUANTITATIVE RESULTS (%) OF ABLATION STUDY FOR EACH KEY COMPONENT IN SCAFM. HSA: HORIZONTAL SPATIAL COMPRESSION ATTENTION; VSA: VERTICAL SPATIAL COMPRESSION ATTENTION; AVP: AVERAGE POOLING; MAP: MAX POOLING. THE BEST RESULTS ARE INDICATED IN BOLD

Method	HSA	VSA	AvP	MaP	F1-Score(%)	IoU(%)
SCAFM-V1	✓	✓	✓	✓	**67.34**	**50.75**
SCAFM-V2			✓	✓	66.48	49.79
SCAFM-V3	✓	✓			66.56	49.88
SCAFM-V4	✓		✓	✓	66.76	50.11
SCAFM-V5		✓	✓	✓	66.61	49.94

Overall, these experimental results comprehensively demonstrate the rationality and necessity of each internal design choice within the SCAFM. The proposed combination of the dual-axis spatial compression attention and the complementary pooling strategies significantly enhances the multimodal contextual reasoning and structural representation capability, thereby improving the segmentation accuracy and the robustness for the detection of ultrathin TLs.

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TABLE VII
IMPACT OF DIFFERENT MODALITY INPUTS UNDER VARYING ENVIRONMENTAL CONDITIONS

Methods	Day		Night		Fog		Snow	
	F1-Score (%)	IoU (%)	F1-Score (%)	IoU (%)	F1-Score (%)	IoU (%)	F1-Score (%)	IoU (%)
RGB only	62.39	45.34	59.09	41.93	59.88	42.73	60.20	43.06
TIR only	61.35	44.25	60.88	43.76	61.23	44.12	61.28	44.18
RGB + TIR (Ours)	**66.94**	**50.31**	**64.24**	**47.32**	**65.90**	**49.14**	**65.45**	**48.64**

4) Ablation Study on Modality Contribution: In complex agricultural environments, relying on a single modality is often insufficient for the robust detection of extremely thin targets. To assess the individual contribution of each modality and to verify whether the performance improvement arises from genuine cross-modal complementarity, an ablation study is conducted using RGB-only, TIR-only, and RGB-TIR fusion inputs under varying environmental conditions, including different illumination levels and adverse weather scenarios (i.e., daytime, nighttime, fog, and snow). Specifically, after combining the simulated and real-world data, the number of samples for each weather condition was adjusted to be identical, resulting in 550 samples per environmental subset. For the RGB-only and TIR-only settings, all modality-interaction and multimodal fusion modules are removed accordingly. To ensure a strictly fair and comparable evaluation, these single-modality baselines are trained under exactly the same optimization protocols, hyperparameters, and data augmentation strategies as the multi-modal fusion framework. The quantitative results are presented in Table VII.

As shown in Table VII, the RGB-TIR fusion model consistently outperforms the single-modality variants across all evaluation metrics. Specifically, under well-lit daytime conditions, the RGB modality achieves competitive performance due to sufficient visual and texture information. However, the performance of the RGB modality is highly sensitive to environmental variations, deteriorating significantly under nighttime, foggy, and snowy conditions owing to inadequate illumination and reduced visibility. In contrast, the TIR modality demonstrates remarkable robustness, maintaining relatively stable performance across all illumination and weather conditions. Notably, the fused RGB-TIR model achieves the best performance across all four challenging scenarios. This consistent superiority indicates that the observed improvement is not dominated by either modality alone but genuinely arises from the complementary characteristics of both modalities. These results confirm that the proposed model fully exploits cross-modal complementarity, leveraging the TIR modality to provide robust structural and thermal cues to counter environmental interference and the RGB modality to provide fine-grained details, thereby substantially enhancing segmentation accuracy and overall robustness in detecting extremely thin TLs.

V. CONCLUSION

In this article, we present AgriSafeNet, a novel RGB-TIR TLD framework specifically designed for smart agricultural applications. AgriSafeNet adopts a multi-branch architecture to effectively integrate fine-grained local features with global contextual representations, thereby significantly improving detection accuracy and structural completeness in complex farmland environments. To address the inherent challenges of multimodal contextual reasoning, we design the MFDFM. By incorporating the FDFB, the MFDFM effectively aligns shallow contextual information and discriminates multiscale features across different modalities. Furthermore, the SCAFM is introduced to explicitly model spatial dependencies, thereby refining the identification of TLs under diverse agricultural conditions. Moreover, the CGFEM dynamically enhances target features by modeling complex semantic relationships between TLs and background regions. This mechanism substantially improves feature discriminability and robustness against interference from visually similar structures, such as crop rows and field boundaries. Beyond the algorithmic contributions, we construct and publicly release AgriSafe-TLD, a comprehensive RGB-TIR dataset designed to support the development and evaluation of multimodal algorithms in agricultural scenarios. Extensive experiments based on the AgriSafe-TLD and VITLD datasets demonstrate that AgriSafeNet effectively suppresses background interference and consistently outperforms state-of-the-art methods in terms of detection accuracy and generalization capability.

In future work, this research will be extended in several directions. First, the proposed framework will be applied to additional linear infrastructure detection tasks. Second, further efforts will focus on improving the robustness of the model under challenging environmental conditions, including extreme weather events and severe seasonal variations. Finally, the integration of the proposed approach with generalized multimodal representations or emerging foundation models will be explored to further improve detection accuracy and zero-shot generalization capability.

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